Arithmetic and harmonic realizations of the Clebsch cubic

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Abstract

We determine the rational square class of Hitchin's degree-two incidence cover of harmonic cubics. Its function field is obtained by adjoining $\sqrt{5J_0}$. We define the Clebsch chart over $\mathbf{Q}(\sqrt{5})$, where $J_0 = 16\sigma_3^2$, and prove that the two configurations over the rational cubic xyz form the complete reduced fibre with residue algebra $\mathbf{Q}(\sqrt{5})$. Their exchanger has nonsquare spinor class after reduction modulo 11. Independently, the degree-six zonal harmonics on the ten icosahedral face axes contain a rational Clebsch four-space as their Petersen eigenspace, and their normalized spherical cubic restricts exactly as

$$\frac{1}{4\pi}\int_{S^2} F_y^3 = -\frac{784000}{1247103}\sigma_3(y).$$

The two realizations use isomorphic Clebsch modules; no rational or integral identification between their displayed embeddings is asserted.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; spherical harmonics; Gaunt invariant.

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1 Introduction

Let

$$H = \ker\Big(\partial_x^2 + \partial_y^2 + \partial_z^2 : \mathbf{Q}[x,y,z]_3 \longrightarrow \mathbf{Q}[x,y,z]_1\Big)$$

be the rational seven-space of harmonic cubics for $Q(x,y,z) = x^2+y^2+z^2$. We use the polarization pairing on cubics normalized by

$$\langle p, (\, a\cdot x\,)^3\rangle = p(a);$$

its restriction to H is nondegenerate. Hitchin associated to a general $[f] \in \mathbf{P}(H)$ two icosahedral six-axis configurations: equivalently, two Mukai–Umemura isotropic three-planes $U \subset H$ with $f \perp U$ [1, 2]. The incidence variety of pairs $([f], U)$ is therefore generically a degree-two cover of $\mathbf{P}(H)$. We determine its rational square class and then study a separate occurrence of the same Clebsch invariant in degree-six spherical harmonics.

Let J_0 denote Hitchin's irreducible invariant sextic, with the normalization in his invariant calculation. Write

$$V = \{(y_1,\dots,y_5) : \textstyle\sum_i y_i = 0\}, \qquad \sigma_3(y) = \frac{1}{3}\sum_i y_i^3$$

for the Clebsch four-space and its invariant cubic. Here and below, $D(\sigma_3)$ is the principal open set $\{\sigma_3 \neq 0\}$.

Theorem 1.1 (Arithmetic incidence cover)**.** *The function field of Hitchin's incidence cover is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}).$$

For the Clebsch chart $\iota_t : \mathbf{P}(V_{\mathbf{Q}(\sqrt{5})}) \to \mathbf{P}(H_{\mathbf{Q}(\sqrt{5})})$ *defined in Section 2,*

$$\iota_t^* J_0 = 16\sigma_3^2.$$

The chart is not defined over $\mathbf{Q}$*; Galois conjugation carries it to the chart of the conjugate icosahedron. The fibre over the rational point* $[xyz]$ *is nevertheless a complete reduced fibre with residue algebra* $\mathbf{Q}(\sqrt{5})$*. Its Galois exchanger has good reduction at* 11*, where its spinor class is the nontrivial element of*

$$\mathrm{PGL}_2(11)/\,\mathrm{PSL}_2(11).$$

The last sentence concerns only the displayed golden fibre and its exchanger; it is not a good-reduction statement for the global geometric incidence cover.

Theorem 1.2 (Degree-six harmonic restriction)**.** *Let* u_{ij}*,* $1 \le i < j \le 5$*, be the explicitly labelled unit axes through opposite icosahedral faces in* (1)*, and set*

$$F_y(\omega) = \sum_{i<j} (y_i + y_j) P_6(u_{ij} \cdot \omega), \qquad P_6(s) = \frac{231s^6 - 315s^4 + 105s^2 - 5}{16}.$$

The coefficient vectors $(y_i + y_j)_{i<j}$ *form the Petersen* (-2)*-eigenspace, and their zonal harmonics give an injective copy of* V *in* $\mathcal{H}_6$*. With* $\langle f, g\rangle = (4\pi)^{-1} \int_{S^2} fg$*, the spherical cubic on this copy is*

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 \, d\omega = -\frac{784000}{1247103} \sigma_3(y).$$

Hitchin proves the generic degree, identifies the branch sextic, constructs the Clebsch chart, computes its restriction, and classifies the fibre over xyz [1, §§2–4, 7–10] [2, §§4–5]. The arithmetic theorem fixes the remaining rational constant in that square class and computes the spinor class of the explicit specialization. The Mukai–Umemura threefold supplies the original threefold [4]; Hitchin supplies the incidence construction and every formula used here. Dye's finite theorem gives the complementary square-5 criterion for Clebsch hexagons [3]. The abstract equation $z^2 = 5J_{\mathbf{Z}}$ is etale on $D(10J_{\mathbf{Z}})$, but its comparison with the geometric incidence variety is proved only after inverting an unspecified integer.

The harmonic theorem is logically independent of the incidence cover. It places the classical degree-six bond-order observable [6, 7] on the same four-space and computes its exact restriction. The coefficient $-784000/1247103$ has denominator divisible by 11, and this paper constructs no rational identification of the two displayed embeddings and no common integral lattice from which they could be obtained by base change. Their relation is an isomorphism class of four-dimensional A_5-modules and the invariant polynomial σ_3, not a specialization between them.

2 The rational incidence cover

2.1 The Clebsch chart

Put $E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$, and let

$$I_t = \{[0 : t : 1], [0 : t : -1], [1 : 0 : t], [-1 : 0 : t], \\ [t : -1 : 0], [-t : -1 : 0]\} \subset \mathbf{P}^2_E.$$

These are the six axes of the standard icosahedron. Define the linear subspace

$$V_t = \{p \in H_E : p(a) = 0 \text{ for every } [a] \in I_t\} \subset H_E.$$

This is the exact inclusion used below. Hitchin proves that V_t has dimension four and describes it as follows [1, §§2–4]. The rotation group of I_t permutes the five triples of mutually orthogonal symmetry planes. Let $q_1, \ldots, q_5$ be the products of the three linear equations in those triples, with $q_1 = xyz$ and relative scalars chosen so that $\sum_i q_i = 0$. Then

$$\iota_t : V_E = \{(y_i) \in E^5 : \textstyle\sum_i y_i = 0\} \longrightarrow V_t, \qquad (y_i) \longmapsto \textstyle\sum_i y_i q_i$$

is an A_5-equivariant isomorphism. Thus the parameter of $xyz = q_1$ is

$$y^\circ = \frac{1}{5}(4, -1, -1, -1, -1), \qquad \sigma_3(y^\circ) = \frac{4}{25}.$$

With the polarization and sextic normalization fixed in the introduction, Hitchin's invariant calculation gives

$$\iota_t^* J_0 = 16\sigma_3^2, \qquad J_0(xyz) = \left(\frac{16}{25}\right)^2 \text{ [1, §10]}.$$

This chart is defined over E, not over $\mathbf{Q}$. The nontrivial automorphism $t \mapsto 1-t$ carries I_t and V_t to the distinct conjugate icosahedron and its vanishing space; the matrix R in Section 3 gives the corresponding transport. Hence ι_t is not a literal rational inclusion $V \hookrightarrow H$. After base change to E, the pullback of the incidence double cover to $D(\sigma_3)$ is split. Galois conjugation exchanges both the two components and the two conjugate charts.

2.2 A rational incidence model

The function-field calculation requires a rational model, which can be given without descending either Clebsch chart. Let $W = \mathbf{Q}^3$ with the quadratic form Q from the introduction. For $a \in W$, infinitesimal rotation about a acts on H; put

$$\omega_a(p, q) = \langle a \cdot p, q\rangle.$$

These three rational skew forms define a closed subscheme

$$X = \{U \in \mathrm{Gr}(3, H) : \omega_a|_U = 0 \text{ for every } a \in W\}.$$

Hitchin identifies $X_{\mathbf{C}}$ with the Mukai–Umemura threefold [2, §§4–5]; in particular X is geometrically integral and smooth. The incidence scheme used here is

$$\mathcal{I} = \{([f], U) \in \mathbf{P}(H) \times X : \langle f, U\rangle = 0\}.$$

It is a projective bundle over X, is therefore normal and integral, and all its displayed equations are defined over $\mathbf{Q}$. Mukai and Umemura introduced the underlying threefold; Hitchin supplies this Grassmannian model and the incidence calculation [4, 2].

2.3 Proof of Theorem 1.1

Let $\pi : \mathcal{I} \to \mathbf{P}(H)$ be the first projection. Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\mathrm{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is a projective bundle over the integral Mukai–Umemura threefold, hence is integral; its generic function field is therefore a quadratic extension.

Square class from the branch divisor. Write $K = \mathbf{Q}(\mathbf{P}(H))$. In characteristic zero every quadratic extension of K has the form $K(\sqrt{d})$. The prime divisors at which d has odd valuation are exactly the branch divisors. Hitchin identifies the branch hypersurface with the irreducible invariant sextic $J_0 = 0$ [1, §§9–10]. Consequently

$$\operatorname{div}(d/J_0) = 2D$$

for a divisor D on $\mathbf{P}(H)$. The class of D is two-torsion. Since $\operatorname{Pic}(\mathbf{P}(H)) = \mathbf{Z}$, D is principal. Thus

$$d = cJ_0 g^2$$

for some $g \in K^\times$ and a unique $c \in \mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. The incidence extension is therefore $K(\sqrt{cJ_0})$.

The local comparison at xyz. Hitchin's classification gives exactly the two distinct configurations I_t and I_{1-t} over $[xyz]$ [1, §§3, 7–9]. Thus π is quasi-finite over this point. Because π is proper, there is a Zariski neighborhood B of $[xyz]$ on which π is finite. Let $\widetilde{B}$ be the normalization of B in the quadratic generic field $K(\sqrt{cJ_0})$. The normal finite scheme $\pi^{-1}(B)$ is birational to $\widetilde{B}$, hence is isomorphic to it. Moreover $J_0(xyz) \neq 0$, so this normalization is etale after shrinking B. The two distinct geometric points therefore form the complete reduced scheme-theoretic fibre, whose residue algebra is

$$E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5}).$$

Choose a local generator of $\mathcal{O}_{\mathbf{P}(H)}(3)$ at $[xyz]$. The equation of the normalization is then $w^2 = cJ_0 u^2$ for a rational unit u; changing the generator multiplies the right side by a square unit. Since $J_0(xyz) = (16/25)^2$, specialization gives the residue algebra $\mathbf{Q}(\sqrt{c})$. Comparison with the preceding fibre proves $c = 5$ in $\mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. This also explains why evaluating the generic square class at the point is legitimate: the point lies in the finite etale locus and every omitted factor is a unit square there.

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \qquad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 1.1. On $D(10J_{\mathbf{Z}})$ it is finite etale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra does not by itself give an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30. Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N, but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

3 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz. Its two icosahedral six-axis configurations are Galois conjugate over $\mathbf{Q}(\sqrt{5})$; modulo 11 they are rational, and the Galois exchange has nontrivial spinor class.

3.1 The golden fibre

Retain the field E, parameter t, and six-set I_t from Section 2. These six projective points are its axes through opposite vertices. Thus an icosahedral configuration here means the unordered six-set I_t; the incidence cover remembers which of the two conjugate six-sets lies over the cubic.

Proposition 3.1 (Golden fibre)**.** *Over a field of characteristic different from* 2 *and* 5*, the points of I_t lie on $xyz = 0$, have a common nonzero norm, have normalized squared inner product* 1/5*, and form a six-arc. The two configurations are conjugate geometric configurations defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's normalized cover of ordered icosahedra they are the complete fibre over $[xyz]$.

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t + 1$, its squared norm is $t^2 + 1 = t + 2$. The inner product of two distinct representatives is $\pm t$, while

$$(t + 2)^2 = 5(t + 1) = 5t^2.$$

This gives the normalized squared inner product 1/5. The twenty three-point determinants are $\pm 2t$ or $\pm 2t^2$, so none vanishes under the stated hypotheses. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. □

3.2 The exchanger

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \qquad R^2 = \operatorname{diag}(1, -1, -1), \qquad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two ordered icosahedra.

3.3 Reduction modulo eleven

The roots of $t^2 - t - 1$ in $\mathbf{F}_{11}$ are 4 and 8. For the displayed equations for I_t and R therefore reduce directly to two rational configurations I_4, I_8 exchanged by R.

Proposition 3.2 (Spinor specialization)**.** *The reduction of R represents the nontrivial element of*

$$T_{11} = \mathrm{PGL}_2(11)/\,\mathrm{PSL}_2(11).$$

Proof. For $Q(x, y, z) = x^2 + y^2 + z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz-plane,

$$R = s_{e_2} s_{e_2 - e_3},$$

so

$$\theta(R) = Q(e_2)Q(e_2 - e_3) = 2 \quad \text{in } \mathbf{F}_{11}^{\times}/\mathbf{F}_{11}^{\times 2}.$$

The class of 2 is nonsquare modulo 11. The standard identifications

$$\mathrm{SO}_3(11) \cong \mathrm{PGL}_2(11), \qquad \Omega_3(11) \cong \mathrm{PSL}_2(11)$$

identify the spinor quotient with T_{11}, proving the claim. □

Integral boundary. Over a finite field of characteristic different from 2 and 5, the abstract quadratic cover gives

$$\#X_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields is proved only away from an unspecified finite set of primes. This global comparison is separate from the good reduction at 11 of the explicit golden fibre and matrix R; we do not replace the finite exceptional set by $p \neq 2, 5$.

4 The degree-six harmonic realization

Put $\phi = (1+\sqrt{5})/2$. The following vectors have squared norm 3; their projective classes are the ten axes through opposite faces of an icosahedron:

$$\begin{aligned}
&v_{12} = (-1,-1,-1), && v_{13} = (-1,-1,1), && v_{14} = (-1,1,-1), && v_{15} = (-1,1,1),\\
&v_{34} = (0,-\phi^{-1},-\phi), && v_{25} = (0,-\phi^{-1},\phi), && v_{45} = (-\phi^{-1},-\phi,0), &&\\
&v_{23} = (-\phi^{-1},\phi,0), && v_{35} = (-\phi,0,-\phi^{-1}), && v_{24} = (\phi,0,-\phi^{-1}). &&
\end{aligned} \tag{1}$$

We use the unit representatives $u_{ij} = v_{ij}/\sqrt{3}$. Two labels are adjacent when the pairs are disjoint. This is the Petersen graph $KG(5,2)$, and the labeling is geometric because, for distinct axes,

$$(u_{ij}\cdot u_{kl})^2 = \begin{cases} 5/9, & \{i,j\}\cap\{k,l\} = \varnothing,\\ 1/9, & \text{otherwise.}\end{cases}$$

Replacing ϕ by its conjugate only conjugates this labeled configuration; the unoriented face-axis set and all formulas below are unchanged up to relabeling.

The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \qquad L(a)(\omega) = \sum_{i<j} a_{ij} P_6(u_{ij}\cdot\omega),$$

where

$$P_6(s) = \frac{1}{16}\left(231s^6 - 315s^4 + 105s^2 - 5\right).$$

We equip $\mathcal{H}_6$ with the normalized spherical inner product

$$\langle f, g\rangle = \frac{1}{4\pi}\int_{S^2} f(\omega)g(\omega)\,d\omega.$$

4.1 Proof of Theorem 1.2

Let A be the Petersen adjacency matrix in the labeling (1). Direct evaluation gives the reproducing-kernel matrix

$$K_{ef} = P_6(u_e\cdot u_f) = \frac{1}{243}(196I + 47J - 112A).$$

The addition theorem for spherical harmonics gives

$$G_{ef} = \langle P_6(u_e\cdot -), P_6(u_f\cdot -)\rangle = \frac{1}{13}K_{ef}.$$

Thus G, not K, is the Gram matrix for the stated inner product. Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1}\oplus V_4\oplus V_5$ are respectively

$$\frac{110}{1053}, \qquad \frac{140}{1053}, \qquad \frac{28}{1053}.$$

They are positive, so L is injective.

The Petersen eigenvalues are $3, -2, 1$ on $\mathbf{1}, V_4, V_5$. For $a_{ij} = y_i + y_j$ with $\sum_i y_i = 0$,

$$(Aa)_{ij} = \sum_{\{k,l\}\cap\{i,j\}=\varnothing} (y_k + y_l) = -2(y_i + y_j).$$

These vectors form the four-dimensional (-2)-eigenspace.

The cubic integral is A_5-invariant on V_4. The invariant cubic line is generated by σ_3, so one nonzero value determines the scalar. At $y = (4, -1, -1, -1, -1)$, exact spherical moments give

$$\frac{1}{4\pi}\int F_y^2 = \frac{2800}{351}, \qquad \frac{1}{4\pi}\int F_y^3 = -\frac{15680000}{1247103},$$

while $\sigma_3(y) = 20$. This proves the coefficient in Theorem 1.2. The moments follow by expanding P_6 and applying

$$\frac{1}{4\pi}\int_{S^2} \omega_1^{2a}\omega_2^{2b}\omega_3^{2c}\,d\omega = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2a+2b+2c+1)!!};$$

monomials with an odd exponent integrate to zero.

Remark 4.1. In the orthonormal Condon–Shortley convention, write $F = \sum_{m=-6}^{6} q_{6m}Y_{6m}$ and

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1}q_{6m_2}q_{6m_3}.$$

The Gaunt formula and $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix}\right) = -20/\sqrt{46189}$ give

$$\int_{S^2} F^3\,d\omega = -\frac{130}{\sqrt{3553\pi}}\,W_6(F).$$

This is the standard unnormalized degree-six bond-order invariant [6, 7, 8].

5 Reproducibility and data availability

The square-class argument and both specialization propositions are proved in the text from Hitchin's incidence calculation and the displayed equations. A paper-local exact-arithmetic bundle independently checks the golden configurations, the exchanger, and its spinor decomposition.

A second paper-local bundle reconstructs the ten axes over $\mathbf{Q}(\sqrt{5})$, distinguishes the reproducing kernel from the normalized spherical Gram matrix, verifies the Petersen labeling, and evaluates the quadratic and cubic moments. Its independent replay uses a separate dense polynomial representation and checks the Wigner $3j$ conversion. Neither calculation uses floating-point arithmetic.

The claim/evidence map, exact programs, canonical certificates, checksums, toolchain requirements, and aggregate replay command are included in the `verification` directory distributed with the paper. The aggregate runner also checks the four theorem-like statements against their frozen identity, verifies label-level correspondence with every trust row, checks a declared packaging allowlist, and requires a warning-free manuscript build. The manifest explicitly claims no Lean coverage: the incidence model, square class, local fibre comparison, spinor specialization, face-axis geometry, and spherical moments are not formal dependencies. The proofs do not depend on availability of the exact-arithmetic audits. The complete source and verification bundle accompanies this manuscript.

6 Conclusion

Hitchin's two conjugate Clebsch charts live over $\mathbf{Q}(\sqrt{5})$, where the branch sextic becomes $16\sigma_3^2$. Their common rational point $[xyz]$ lies in the finite etale locus of the rational incidence model. Its complete reduced fibre determines the remaining rational square class. The explicit configurations also have a well-defined specialization at 11, without requiring a global integral incidence model there.

The degree-six face-axis construction gives a second exact role for σ_3: it is the restriction of the normalized spherical cubic to the Petersen four-space. The two constructions use isomorphic Clebsch modules and the same invariant polynomial, but their displayed embeddings are not identified over $\mathbf{Q}$, and no integral comparison is asserted.

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